

Route Clearing

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1. Experiment Purpose
2. Hardware Requirements
3. Implementation Principle
4. Experiment Steps
5. Experiment Phenomenon

1. Experiment Purpose

Previously, we learned how to read the data of the four-way tracking sensor to implement the line-following function. In this section, on top of that, we will combine ultrasonic and servo control to implement the line tracking and obstacle clearing function.

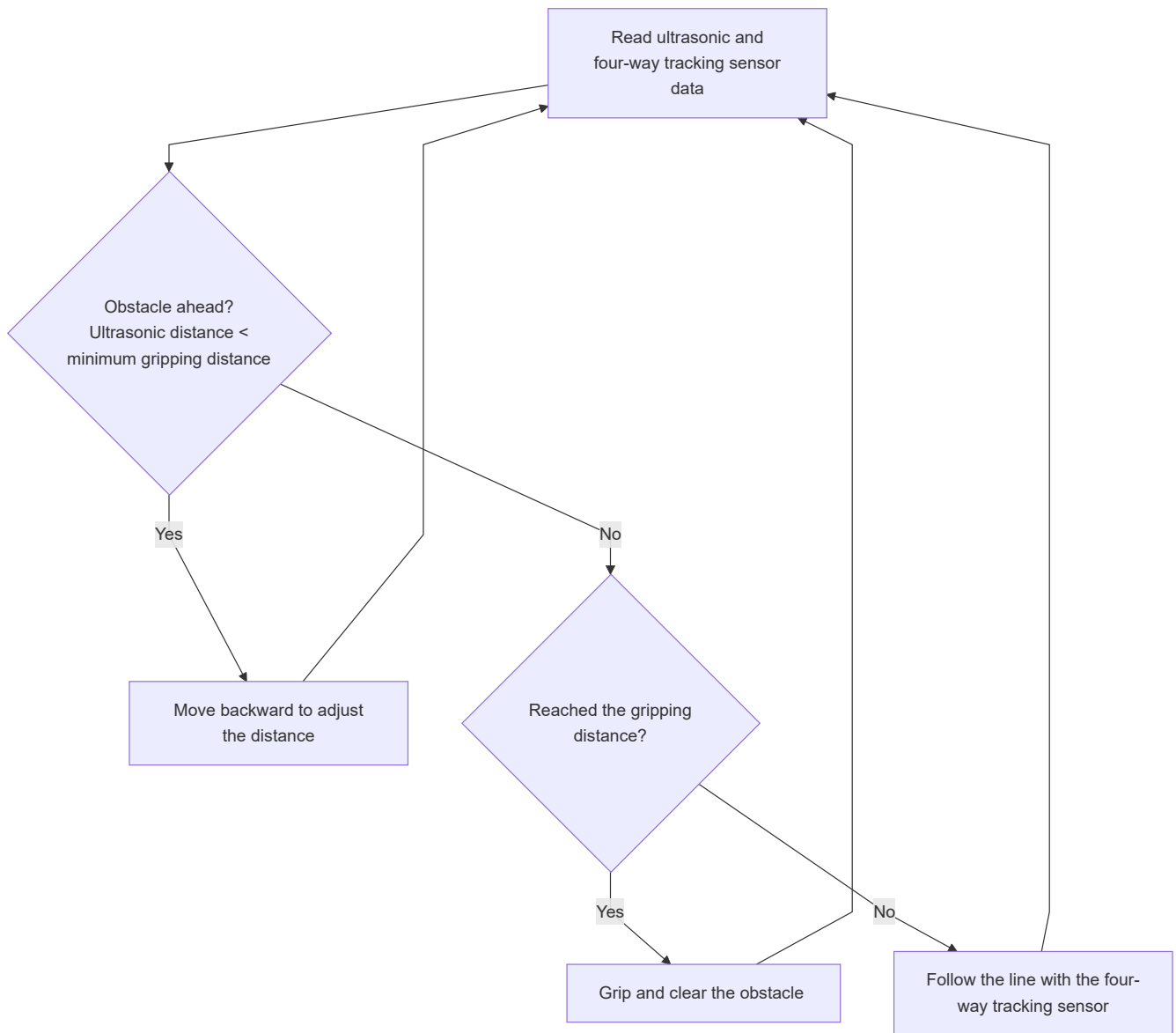
2. Hardware Requirements

Make sure the battery is fully charged and the four-way tracking sensor has been adjusted to a suitable sensitivity with the potentiometer.

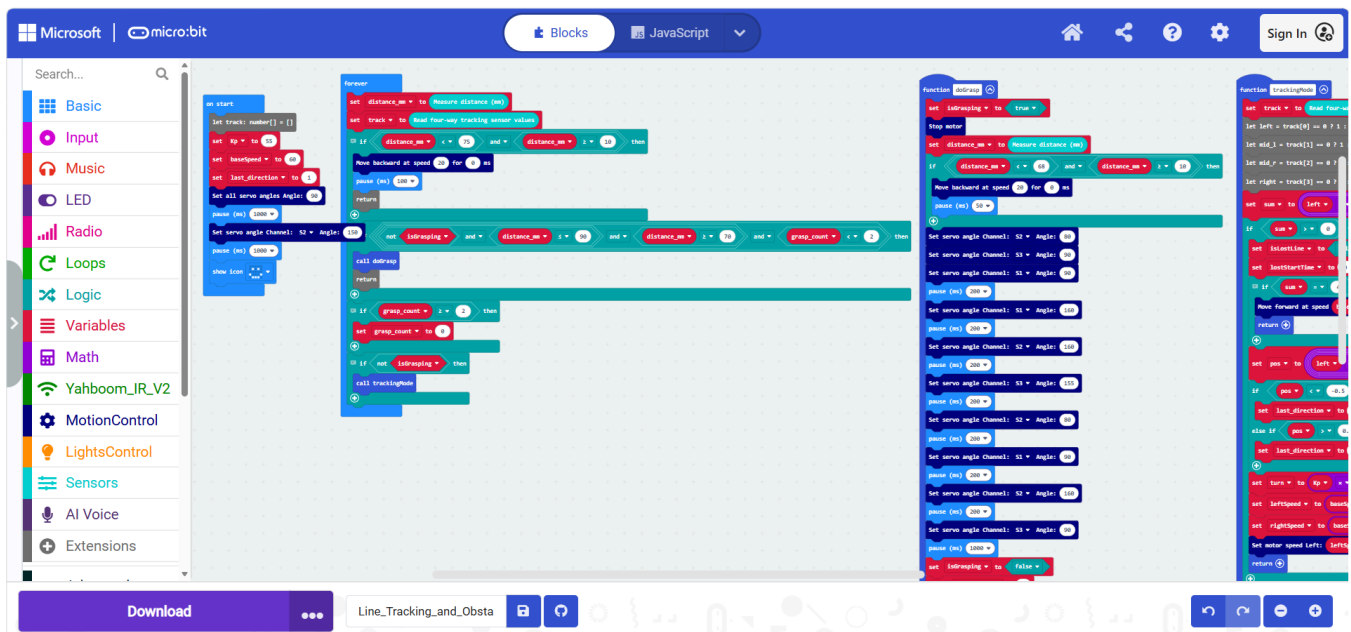
Lay the track flat on the ground.

3. Implementation Principle

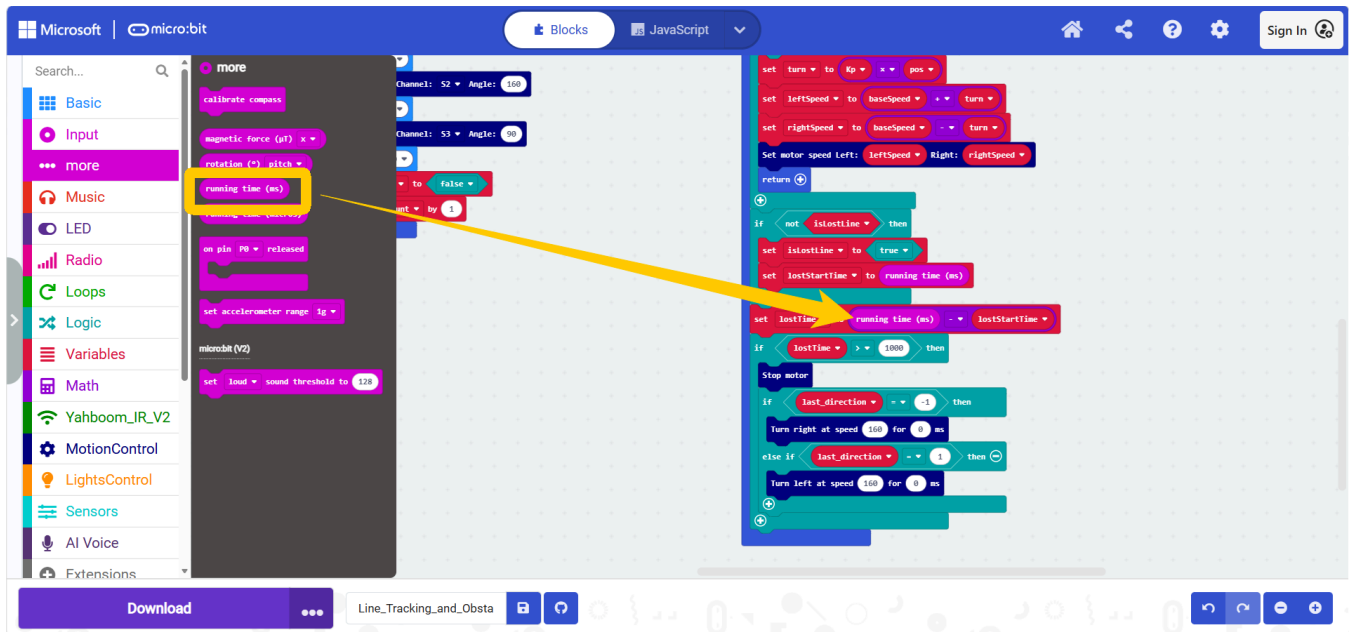
Read the data from the four-way tracking sensor, determine the state of the car's position on the line based on the data, and apply different controls for different states.



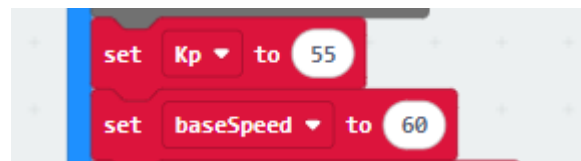
The program is as follows.



Running time (ms) is located in the `more` subcategory of the `Input` category.



You can adjust the following parameters to set the car's driving speed and turning amplitude.



If you need to adjust the gripping speed, you can modify the pause (ms) parameter.



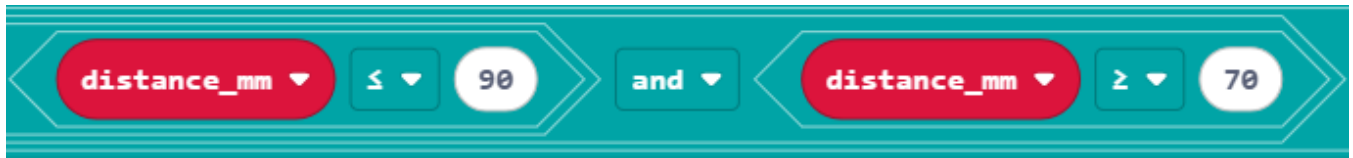
4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable between the computer and the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `22.Route_Clearing.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. After flashing, unplug the data cable and place the car on the track to be followed.
4. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

The car will follow the track to do line-following. When the ultrasonic sensor detects the distance to the EVA block and the distance reaches the gripping range, the car will stop and clear the block. Place the block at the front-left of the car (when the obstacle is close, the car will move backward to adjust the distance). After the ultrasonic sensor detects no obstacle, the car continues line-following.

If you need to adjust the gripping distance range, you can modify the following parameters.



It is recommended to place the EVA obstacle block on the straight section of the track after the car turns; the ultrasonic distance measurement is more accurate on this section.

